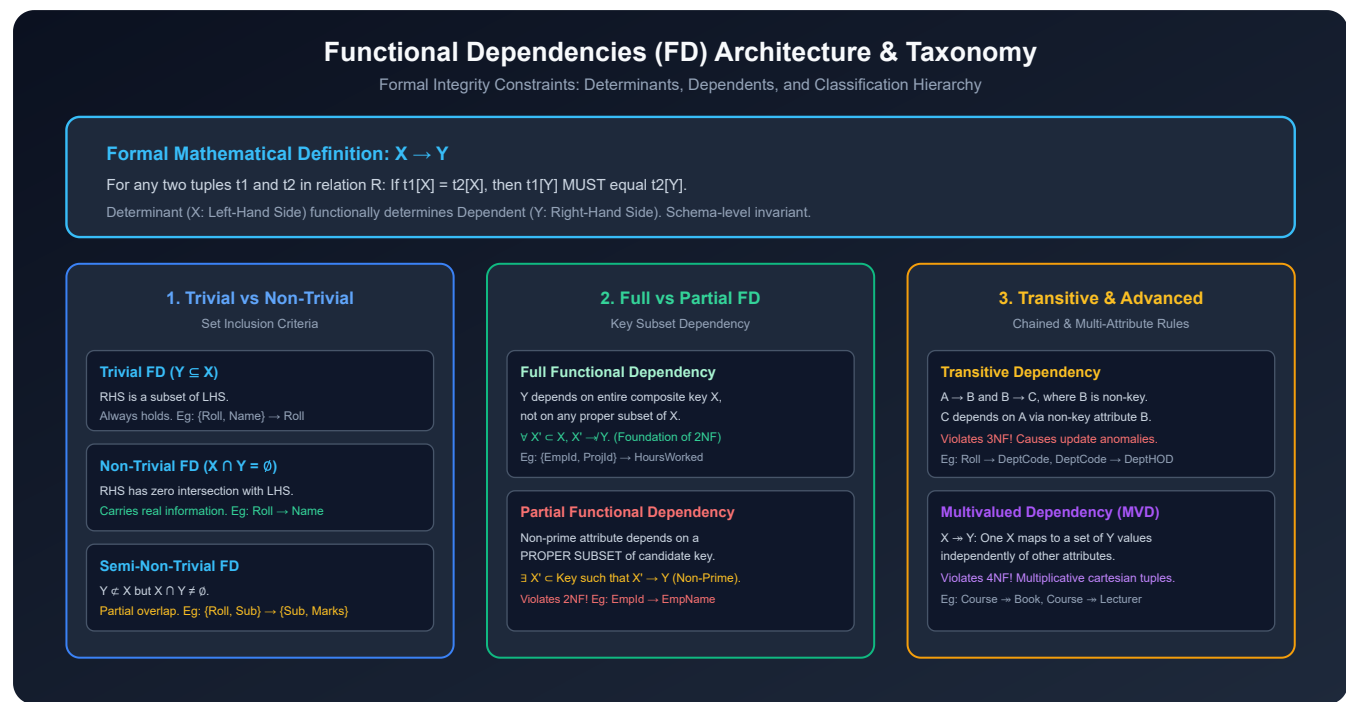


Module 01: Functional Dependencies & Types



> **Course:** Database Management Systems (AKTU Code: **BCS501**)

****Topic Focus:**** Formal Definition of FD, Determinant vs Dependent, Schema Constraints vs Instance Data, Trivial, Non-Trivial, Semi-Trivial, Full vs Partial FDs.

1. Functional Dependency (FD) Core Concept & Philosophy

Database design mein Functional Dependency relation ke attributes ke beech ka ek **semantic integrity constraint** hota hai. Yeh decide karta hai ki ek table ke andar values kis tarah interrelated hain aur redundancy kahan paida ho rahi hai.

1.1 Formal Mathematical Definition

Ek relation schema $R(A_1, A_2, \dots, A_n)$ par, jisme $X \subseteq R$ aur $Y \subseteq R$ attributes ke sets hain:

$$X \rightarrow Y$$

$$\text{if } t_1[X] = t_2[X] \text{ then } t_1[Y] = t_2[Y]$$

Yeh Functional Dependency tabhi valid maani jaati hai jab relation R ke kisi bhi valid legal instance mein:

$$t_1[X] = t_2[X] \implies t_1[Y] = t_2[Y]$$

$$\text{forall } t_1, t_2 \in R, \quad \text{if } t_1[X] = t_2[X] \implies t_1[Y] = t_2[Y]$$

Bilingual Explanation: Agar do tuples (rows) t_1 aur t_2 ke paas attribute set X ki value exactly same hai, toh unke attribute set Y ki value bhi compulsory roop se **same honi chahiye!**

Determinant (X): Left-Hand Side (LHS) attribute jo decide karta hai.

Dependent (Y): Right-Hand Side (RHS) attribute jo decide hota hai.

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LHS: X	—————>	RHS: Y
(Determinant)		(Dependent)

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Eg: Roll_No Eg: Student_Name, Branch

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Notice:

****Schema Constraint vs Instance Property:****

Functional Dependency table ke current snapshot ya instance ko dekh kar final nahi ki ja sakti! Yeh ****real-world business logic aur schema rules**** par depend karti hai.

Instance me do alag students ka naam same ho sakta hai ya coincidence se kuch rows match kar sakti hain, par FD schema level rule hota hai jo har bhavishya (future) ke instance par bhi hold karna chahiye.

2. Comprehensive Taxonomy of Functional Dependencies

Functional Dependencies ko unke attribute relationship aur mathematical properties ke basis par classify kiya jaata hai:

2.1 Trivial Functional Dependency

- **Condition:** $Y \subseteq X$ (RHS is a subset of LHS).
- **Explanation:** Jab right side ka attribute already left side ke attribute set ka part ho.
- **Validity:** Yeh har relation instance par **hamesha true** hoti hai, bina kisi data ko check kiye!
- **Examples:**

- $\{A, B\}$

$\rightarrow A$

- $\{Roll_No, Name\}$

$\rightarrow Roll_No$

- A

$\rightarrow A$

2.2 Non-Trivial Functional Dependency

- **Condition:** $X \cap Y = \emptyset$ (LHS and RHS have zero common attributes).
- **Explanation:** Jab right side ka koi bhi attribute left side me present nahi hota.
- **Significance:** Real-world database design me Non-Trivial FDs hi meaningful information aur constraints capture karti hain!
- **Examples:**

- $Roll_No$

$\rightarrow Student_Name$

- Emp_ID

$\rightarrow Salary$

- $\{Dept_ID, Project_ID\}$

$\rightarrow Budget$

2.3 Semi-Non-Trivial (Semi-Trivial) FD

• **Condition:** Y

$\subseteq X$ and $X \cap Y$

$\emptyset$.

• **Explanation:** Jab RHS ka kuch part LHS me common ho, par RHS poori tarah LHS ka subset na ho.

• **Example:**

- $\{A, B\}$

$\rightarrow \{B, C\}$

- By Decomposition Rule: $\{A, B\}$

$\rightarrow B$ (Trivial) and $\{A, B\}$

$\rightarrow C$ (Non-Trivial).

3. Full vs Partial Functional Dependencies (Gateway to 2NF)

Yeh distinction Normalization ke **Second Normal Form (2NF)** ka core pillar hai!

Parameter	Full Functional Dependency	Partial Functional Dependency
Formal Definition		

$\rightarrow Y$ holds, and X

forall $X' \subseteq X$, $X' \rightarrow Y$

or

$\rightarrow Y \mid \exists X' \subseteq X \text{ such that } X' \rightarrow Y$

$\rightarrow Y \mid$

Simple Meaning	Y poore composite key par depend karta hai, kisi akele tukde par nahi.	Y composite key ke kisi ek chote hisse (proper subset) par depend kar jata hai.
Candidate Key Type	Composite Candidate Key required	Composite Candidate Key required
Impact on Database	Normalization ke liye desirable (2NF satisfied)	Redundancy paida karta hai (**Violates 2NF!**)
Real Example		

$\rightarrow Hours_Worked \mid \{Emp_ID, Project_ID\}$

$\rightarrow Emp_Name$ (Emp_ID

$\rightarrow Emp_Name$)

4. Transitive Functional Dependency (Gateway to 3NF)

Definition: Jab ek non-key attribute kisi doosre non-key attribute par functionally depend karta hai through an indirect chain:

X

$Y \twoheadrightarrow Z$ and $Z \twoheadrightarrow X$

$Z \twoheadrightarrow X$

$X \twoheadrightarrow Z$

Where X is candidate key, but Y is NOT a super key and Z is a non-prime attribute.

Real-World Pitfall:

- Roll No

$\text{Department_Code} \twoheadrightarrow \text{Department_Head}$

- Department Code

$\text{Department_Head} \twoheadrightarrow \text{Roll_No}$

- Yahan Roll No

$\text{Department_Head} \twoheadrightarrow \text{Roll_No}$ indirectly transitive hai. Agar department head change hoga, toh har student ki row me update karna padega (Update Anomaly).

- Yeh dependency **Third Normal Form (3NF)** ko violate karti hai!

5. Architectural Diagram

6. Summary Comparison Table for Quick Revision

FD Type	Mathematical Form	Always Valid?	Violates Normal Form?	Real-Life Example
Trivial	$Y \subseteq X$	Yes (Trivially)	None	

$A \twoheadrightarrow B$

Non-Trivial	$X \cap Y = \emptyset$	Depends on schema	Core constraint
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$\text{Emp_Salary} \twoheadrightarrow \text{Emp_Name}$

Semi-Trivial

$X \subseteq Y$ and $X \cap Y \neq \emptyset$

$\emptyset \twoheadrightarrow A$ | Depends on schema | Can be decomposed | $\{A, B\}$

$\{B, C\} \twoheadrightarrow A$

Partial

$\rightarrow \text{ext}\{\text{Non-Prime}\} \mid \text{Depends on schema} \mid \textbf{Violates 2NF} \mid \Emp_ID

$\rightarrow \text{Emp_Name} \mid \text{in Project allocation} \mid$

Transitive

$\rightarrow X$

$\rightarrow Y \mid \text{Depends on schema} \mid \textbf{Violates 3NF} \mid \Student

$\rightarrow \text{Dept}$

$\rightarrow \text{HOD} \mid$

Multivalued	$X \twoheadrightarrow Y$	Independent multi-facts	Violates 4NF	$\text{Course} \twoheadrightarrow \text{Books}$
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